

```
[2]: # Takes the dataset and uses the rocket column to call the API and append the data to the list
def getBoosterVersion(data):
    for x in data['rocket']:
        if x:
            response = requests.get("https://api.spacexdata.com/v4/rockets/"+str(x)).json()
            BoosterVersion.append(response['name'])
```

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[3]: # Takes the dataset and uses the launchpad column to call the API and append the data to the list
def getLaunchSite(data):
    for x in data['launchpad']:
        if x:
            response = requests.get("https://api.spacexdata.com/v4/launchpads/"+str(x)).json()
            Longitude.append(response['longitude'])
            Latitude.append(response['latitude'])
            LaunchSite.append(response['name'])
```

```
[4]: # Takes the dataset and uses the payloads column to call the API and append the data to the lists
def getPayloadData(data):
    for load in data['payloads']:
        if load:
            response = requests.get("https://api.spacexdata.com/v4/payloads/"+load).json()
            PayloadMass.append(response['mass_kg'])
            Orbit.append(response['orbit'])
```



```
[5]: # Takes the dataset and uses the cores column to call the API and append the data to the lists
def getCoreData(data):
    for core in data['cores']:
        if core['core'] != None:
            response = requests.get("https://api.spacexdata.com/v4/cores/"+core['core']).json()
            Block.append(response['block'])
            ReusedCount.append(response['reuse_count'])
            Serial.append(response['serial'])
        else:
            Block.append(None)
            ReusedCount.append(None)
            Serial.append(None)
        Outcome.append(str(core['landing_success'])+' '+str(core['landing_type']))
        Flights.append(core['flight'])
        GridFins.append(core['gridfins'])
        Reused.append(core['reused'])
        Legs.append(core['legs'])
        LandingPad.append(core['landpad'])
```



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[6]: spacex_url = "https://api.spacexdata.com/v4/launches/past"
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response = requests.get(spacex_url)  
print(response.content)
```

```
b<!DOCTYPE html>\n<!--[if lt IE 7]> <html class="no-js ie6 oldie" lang="en-US"> <![endif-->\n<!--[if IE 7]> <html class="no-js ie7 oldie" lang="en-US"> <![endif-->\n<!--[if IE 8]> <html class="no-js ie8 oldie" lang="en-US"> <![endif-->\n<!--[if gt IE 8]><!--> <html class="no-js" lang="en-US"> <!--<![endif-->\n<head>\n<title>spacexdata.com | 525: SSL handshake failed</title>\n<meta charset="UTF-8" /\n<meta http-equiv="Content-Type" content="text/html; charset=UTF-8" /\n<meta http-equiv="X-UA-Compatible" content="IE=Edge" /\n<meta name="robots" content="noindex, nofollow" /\n<meta name="viewport" content="width=device-width,initial-scale=1" /\n<link rel="stylesheet" id="cf_styles-css" href="/cdn-cgi/styles/main.css" /\n</head>\n<body>\n<div id="cf-wrapper">\n  <div id="cf-error-details" class="p-0">\n    <header class="mx-auto pt-10 lg:pt-6 lg:px-8 w-240 lg:w-full mb-8">\n      <h1 class="inline-block sm:block sm:mb-2 font-light text-60 lg:text-4xl text-black-dark leading-tight mr-2">\n        <span class="inline-block">SSL handshake failed</span>\n        <span class="code-label">Error code 525</span>\n      </h1>\n    </div>\n    Visit <a href="https://www.cloudflare.com/5xx-error-landing?utm_source=errorcode_525&utm_campaign=api.spacexdata.com" target="_blank" rel="noopener noreferrer">cloudflare.com</a> for more information.\n    </div>\n    <div class="mt-3">2026-09-07 15:15:52 UTC</div>\n  </div>\n  <div class="my-8 bg-gradient-gray">\n    <div class="w-240 lg:w-full mx-auto">\n      <div class="clearfix md:px-8">\n        <div id="cf-browser-status" class="relative w-1/3 md:w-full py-15 md:p-0 md:py-8 md:text-left md:border-solid md:border-0 md:border-b md:border-gray-400 overflow-hidden float-left md:float-none text-center">\n          <div class="relative mb-10 md:m-0">\n            <span class="cf-icon-browser block md:hidden h-20 bg-center bg-no-repeat"></span>\n            <span class="cf-icon-ok w-12 h-12 absolute left-1/2 md:left-auto md:right-0 md:top-0 -ml-6 -bottom-4"></span>\n          </div>\n          <span class="md:block w-full truncate">You</span>\n          <h3 class="md:inline-block mt-3 md:mt-0 text-2xl text-gray-600 font-light leading-1.3">\n            Browser\n          </h3>\n          <span class="leading-1.3 text-2xl text-green-success">Working</span>\n        </div>\n        <div id="cf-cloudflare-status" class="relative w-1/3 md:w-full py-15 md:p-0 md:py-8 md:text-left md:border-solid md:border-0 md:border-b md:border-gray-400 overflow-hidden float-left md:float-none text-center">\n          <div class="relative mb-10 md:m-0">\n            <a href="https://www.cloudflare.com/5xx-error-landing?utm_source=errorcode_525&utm_campaign=api.spacexdata.com" target="_blank" rel="noopener noreferrer">\n              <span class="cf-icon-cloud block md:hidden h-20 bg-center bg-no-repeat"></span>\n              <span class="cf-icon-ok w-12 h-12 absolute left-1/2 md:left-auto md:right-0 md:top-0 -ml-6 -bottom-4"></span>\n            </a>\n          </div>\n          <span class="md:block w-full truncate">Marseille</span>\n          <h3 class="md:inline-block mt-3 md:mt-0 text-2xl text-gray-600 font-light leading-1.3">\n            <a href="https://www.cloudflare.com/5xx-error-landing?utm_source=errorcode_525&utm_campaign=api.spacexdata.com" target="_blank" rel="noopener noreferrer">\n              Cloudflare\n            </a>\n          </h3>\n          <span class="leading-1.3 text-2xl text-green-success">Working</span>\n        </div>\n        <div id="cf-host-status" class="cf-error-source relative w-1/3 md:w-full py-15 md:p-0 md:py-8 md:text-left md:border-solid md:border-0 md:border-b md:border-gray-400 overflow-hidden float-left md:float-none text-center">\n          <div class="relative mb-10 md:m-0">\n            <span class="cf-icon-server block md:hidden h-20 bg-center bg-no-repeat"></span>\n            <span class="cf-icon-error w-12 h-12 absolute left-1/2 md:left-auto md:right-0 md:top-0 -ml-6 -bottom-4"></span>\n          </div>\n          <span class="md:block w-full truncate">api.spacexdata.com</span>\n          <h3 class="md:inline-block mt-3 md:mt-0 text-2xl text-gray-600 font-light leading-1.3">\n            Host\n          </h3>\n          <span class="leading-1.3 text-2xl text-red-error">Error</span>\n        </div>\n      </div>\n      <div class="w-240 lg:w-full mx-auto mb-8 lg:px-8">\n        <div class="clearfix">\n          <div class="w-1/2 md:w-full float-left pr-6 md:pb-10 md:pr-0 leading-relaxed">\n            <h2 class="text-3xl font-normal leading-1.3 mb-4">What happened?</h2>\n            <p>Cloudflare is unable to establish an SSL connection to the origin server.</p>\n          </div>\n          <div class="w-1/2 md:w-full float-left leading-relaxed">\n            <h2 class="text-3xl font-normal leading-1.3 mb-4">What can I do?</h2>\n            <p>\n              <h3 class="text-15 font-semibold mb-2">If you're a visitor of this website:</h3>\n              <p class="mb-6">Please try again in a few minutes.</p>\n              <h3 class="text-15 font-semibold mb-2">If you're the owner of this website:</h3>\n              <p><span>It appears that the SSL configuration used is not compatible with Cloudflare. This could happen for a several reasons, including no shared cipher suites.</span> <a rel="noopener noreferrer" href="https://developers.cloudflare.com/support/troubleshooting/http-status-codes/cloudflare-5xx-errors/error-525/">Additional troubleshooting information here.</a></p>\n            </div>\n          </div>\n        </div>\n      </div>\n      <div class="cf-error-footer cf-wrapper w-240 lg:w-full py-10 sm:py-4 sm:px-8 mx-auto text-center sm:text-left border-solid border-0 border-t border-gray-300">\n        <p class="text-13">\n          <span class="cf-footer-item sm:block sm:mb-1">Cloudflare Ray ID: <strong class="font-semibold">a376abdcff92e1e9</strong></span>\n          <span class="cf-footer-separator sm:hidden">&bull;</span>\n          <span id="cf-footer-item-ip" class="cf-footer-item hidden sm:block sm:mb-1">\n            Your IP:\n            <button type="button" id="cf-footer-ip-reveal" class="cf-footer-ip-reveal-button">Click to reveal</button>\n          </span>\n          <span class="hidden" id="cf-footer-ip">41.238.20.24</span>\n          <span class="cf-footer-separator sm:hidden">&bull;</span>\n          <span>Performance &amp; security by</span> <a rel="noopener noreferrer" href="https://www.cloudflare.com/5xx-error-landing?utm_source=errorcode_525&utm_campaign=api.spacexdata.com" id="brand-link" target="_blank">Cloudflare</a>\n        </p>\n      </div>\n    </div>\n  </div>\n</body>\n</html>
```

```
[7]: static_json_url='https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/API_call_spacex_api.json'
      response=requests.get(static_json_url)
      response.status_code
```

```
[7]: 200
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[11]: # Lets take a subset of our dataframe keeping only the features we want and the flight number, and date_utc.
      data = data[['rocket', 'payloads', 'launchpad', 'cores', 'flight_number', 'date_utc']]

      # We will remove rows with multiple cores because those are falcon rockets with 2 extra rocket boosters and rows that have multiple payloads in a single
      data = data[data['cores'].map(len)==1]
      data = data[data['payloads'].map(len)==1]

      # Since payloads and cores are lists of size 1 we will also extract the single value in the list and replace the feature.
      data['cores'] = data['cores'].map(lambda x : x[0])
      data['payloads'] = data['payloads'].map(lambda x : x[0])

      # We also want to convert the date_utc to a datetime datatype and then extracting the date leaving the time
      data['date'] = pd.to_datetime(data['date_utc']).dt.date

      # Using the date we will restrict the dates of the launches
      data = data[data['date'] <= datetime.date(2020, 11, 13)]
```